



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences



Student Name: Akshitha TR

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Course Code: CSA5115

Slot:A

Course Name: CRYPTOGRAPHY AND NETWORK SECURITY

Course Faculty: Dr.T.Sangeetha

Project Title : “Ransomware Detection and Prevention System”

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description:

Ransomware Prevention and Detection System is a cybersecurity-based project designed to identify and prevent ransomware attacks before they cause significant damage to files and systems. Ransomware is a type of malware that can encrypt or modify important files and make them inaccessible to users.

The proposed system continuously monitors file and system activities to detect suspicious behavior, such as unusual file modifications, rapid file encryption, abnormal file-access patterns, and sudden changes in large numbers of files. When potentially malicious activity is detected, the system generates an alert and can take preventive actions such as blocking the suspicious process and protecting targeted files.

The system aims to provide an additional layer of security by combining real-time monitoring, ransomware behavior detection, alert generation, and preventive mechanisms. It can help reduce data loss and improve the early detection of ransomware attacks. The project is intended to provide a simple and effective security solution for protecting sensitive data in personal and organizational systems.

Student Signature

Guide Signature